EXPEREIMENT - 03

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Multi-Byte Data and Array Operations Using 8086 Indirect Addressing

a) To add two 32-bit (double-precision) numbers using register-indirect and based addressing with SI, DI and BX.

b) To find the sum of N 16-bit numbers stored in a word array with a 32-bit result using a loop.

OBJECTIVE

To perform 32-bit addition using register-indirect and based addressing modes and calculate the sum of N 16-bit numbers stored in a word array using 8086 assembly language programming.

ALGORITHM

1. Initialize the data segment and load the starting memory address into the appropriate index or base registers.
2. Load the multi-byte operands or word array elements using indirect addressing.
3. Perform addition using suitable 8086 arithmetic instructions.
4. Repeat the addition using a loop until all array elements are processed.
5. Store the final result in memory and verify the output.

(a) Addition of Two 32-Bit Numbers

Formula

$$\text{Result} = \text{Number 1} + \text{Number 2}$$

Input

- Number 1 = 12345678H
- Number 2 = 11111111H

Code (8086 Assembly Language)

```
.MODEL SMALL

.STACK 100H

.DATA
```

```
NUM1 DD 12345678H
NUM2 DD 11111111H
RES  DD ?
.CODE
MAIN PROC
   MOV AX, @DATA
   MOV DS, AX
   LEA SI, NUM1
   LEA DI, NUM2
   LEA BX, RES
   MOV AX, [SI]
   ADD AX, [DI]
   MOV [BX], AX
   MOV AX, [SI+2]
   ADC AX, [DI+2]
   MOV [BX+2], AX
   MOV AH, 4CH
   INT 21H
MAIN ENDP
END MAIN
```

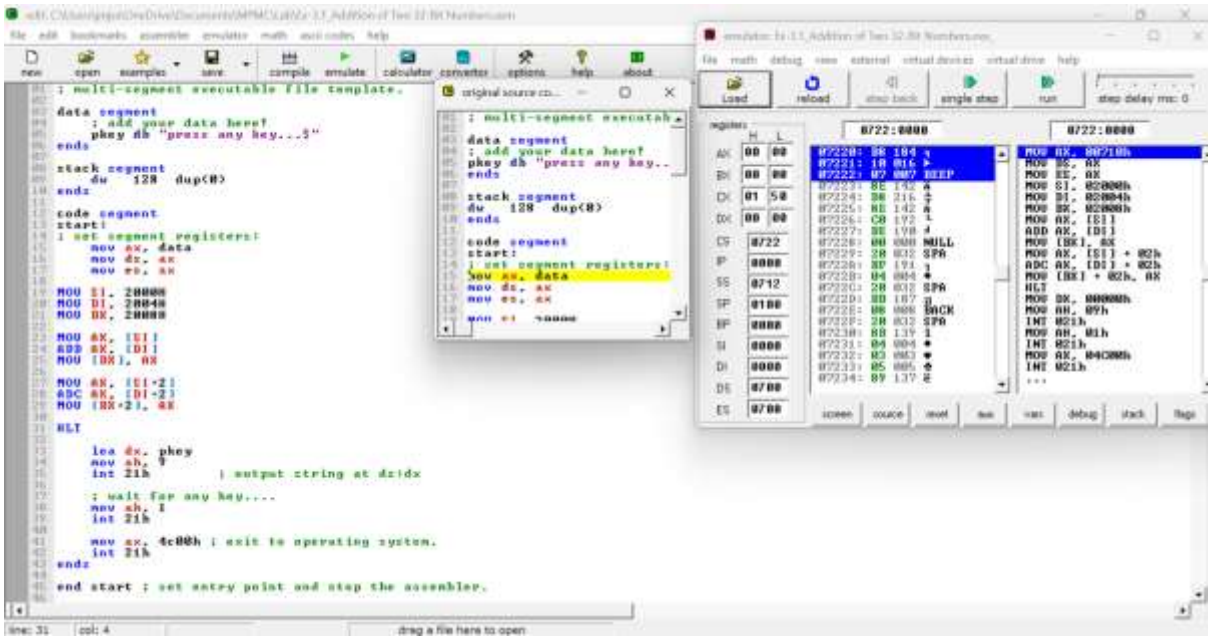
Output

- Number 1 = 12345678H
- Number 2 = 11111111H
- Result = **23456789H**

Result memory:

- RES = 6789H (Low word)
- RES+2 = 2345H (High word)

EXECUTION:



(b) Sum of N 16-Bit Numbers Using a Loop

Formula

$$\text{Sum} = \sum_{i=1}^{N} A_i$$

Input

- N = 4
- Array = 1000H, 2000H, 3000H, 4000H

Code (8086 Assembly Language)

```
.MODEL SMALL
.STACK 100H
.DATA
ARR DW 1000H, 2000H, 3000H, 4000H
N DW 4
SUM DD 0
.CODE
```

```
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX
    LEA SI, ARR
    MOV CX, N
    XOR AX, AX
    XOR DX, DX
NEXT:
    ADD AX, [SI]
    ADC DX, 0
    ADD SI, 2
    LOOP NEXT
    MOV WORD PTR SUM, AX
    MOV WORD PTR SUM+2, DX
    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```

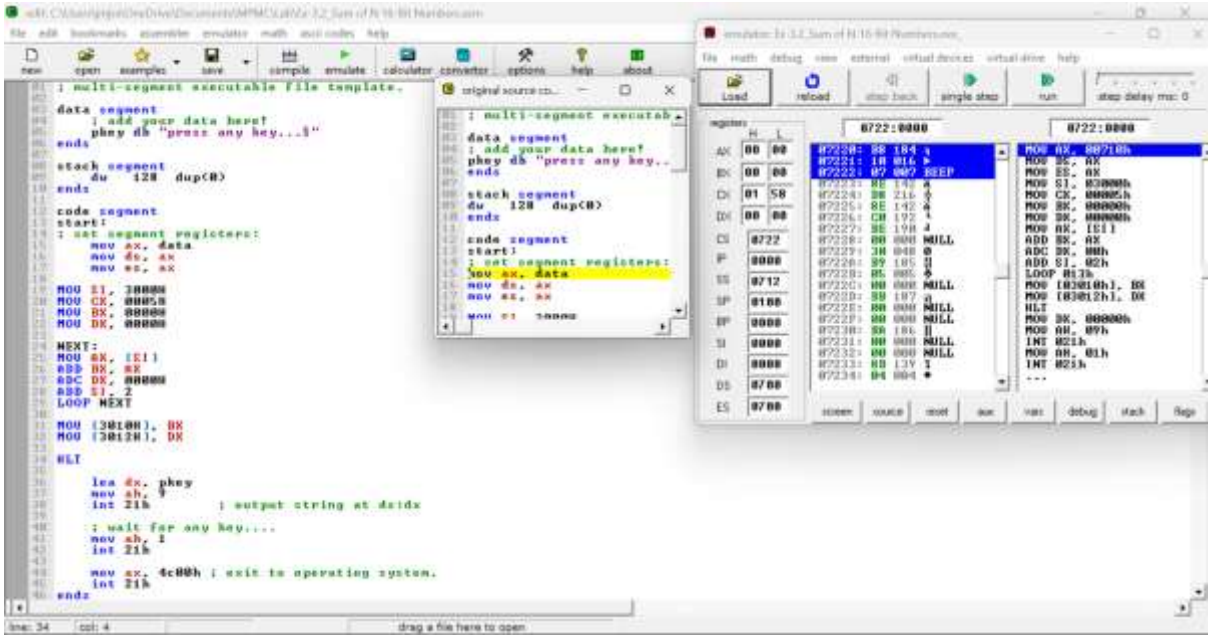
Output

- Array elements = 1000H, 2000H, 3000H, 4000H
- Number of elements = 4
- Sum = **0000A000H**

Result memory:

- SUM = A000H (Low word)
- SUM+2 = 0000H (High word)

EXECUTION:



STUDENT REFLECTION

This experiment helped me understand register-indirect and based addressing modes in the 8086 microprocessor. I learned to add 32-bit numbers, process word arrays using loops, and store multi-byte results in memory.

RESULT

Thus, the 8086 assembly language programs for 32-bit addition using indirect addressing and summation of N 16-bit array elements using a loop were developed and verified successfully.